

KUMARESAN M

Backend Engineer · Node.js · AWS · Distributed Systems

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PROFESSIONAL SUMMARY

Backend engineer with 2+ years building production server-side systems across distributed cloud infrastructure, async pipelines, and IoT edge integration. Designed a DICOM medical imaging pipeline on AWS (S3 + SQS + Lambda) — event-driven, retry-resilient, currently running in production for a healthcare client. Built Python edge services on Raspberry Pi with Bluetooth device management and over-the-air firmware deployment. Shipped multi-tier RBAC APIs on a live e-commerce platform serving customers in Canada. Strong grasp of backend fundamentals: event-driven architecture, message queues, API design, database optimisation, and production observability. Seeking a senior backend engineering role where deep server-side work is the primary focus.

CORE SKILLS

Backend	Node.js · Python · Express.js · FeathersJS · REST API Design · RBAC · WebSockets
Cloud & AWS	S3 · SQS · Lambda · Secrets Manager · Async Pipelines · Event-driven Architecture
Databases	PostgreSQL · MySQL · MongoDB · Firebase
Distributed	Message Queues · Microservices · Background Jobs · Retry Logic · Idempotency
IoT & Edge	Raspberry Pi · Bluetooth · FOTA (OTA Updates) · Linux · DICOM Imaging
Practices	Structured Logging · API Documentation (Swagger) · Agile / Scrum · Git · Postman

WORK EXPERIENCE

Backend Engineer (Full Stack)

Mar 2024 – Present

Rugas Technologies Pvt. Ltd. · Bangalore, KA

RADX.report — Medical Ultrasound Reporting Platform

- Architected end-to-end async DICOM image ingestion pipeline: S3 upload → SQS event trigger → Lambda processor → metadata storage. Chose SQS over direct Lambda invocation for retry resilience, exactly-once delivery semantics, and decoupled scaling between ingestion and processing layers.
- Built Python background service on Raspberry Pi: Bluetooth device pairing with medical hardware, secure REST cloud sync, structured logging, and FOTA firmware deployment — enabling zero-touch remote updates across production devices without on-site intervention.
- Designed FeathersJS service layer with clean separation of concerns, consumed by React frontend for DICOM-based radiologist reporting workflows; secured credentials via AWS Secrets Manager.
- Pipeline passed remote healthcare compliance audit and has been running in production — zero downtime incidents attributable to the pipeline architecture since deployment.

KarmaTickets.ca — Canada-based E-commerce Ticketing Platform

- Owned production backend for a live Canada-based ticketing platform: JWT-based authentication, session management, and order processing pipelines using Node.js and FeathersJS.
- Designed multi-tier RBAC from scratch — role definitions, permission middleware, and enforcement at both route and data-access layer — eliminating privilege escalation vectors across admin, organiser, and buyer roles.
- Reduced mean time to resolve production incidents from hours to minutes by implementing structured logging and observability across Node.js backend and Linux-based device environments.

- ▶ Resolved critical production bugs beyond assigned scope, partnering directly with QA and product stakeholders — recognised with Certificate of Appreciation for reliability and commitment.

KEY PROJECTS

Async DICOM Ingestion Pipeline

Production · 2024

Node.js · AWS S3 · SQS · Lambda · Python

- ▶ Designed fully async backend pipeline: DICOM images → S3 → SQS event trigger → Lambda processor → structured metadata storage. Chose SQS over direct Lambda for retry resilience, idempotent processing, and decoupled scaling.
- ▶ Running in production with zero manual intervention. Successfully passed remote compliance audit by a healthcare client.

Python IoT Edge Service + FOTA

Production · 2024

Python · Raspberry Pi · Bluetooth · Linux

- ▶ Built Python background daemon on Raspberry Pi: Bluetooth device pairing, secure REST cloud sync, structured logging, and FOTA firmware deployment — zero-touch remote updates, no on-site visits required.

Multi-tier RBAC API — KarmaTickets.ca

Production · 2024

Node.js · FeathersJS · PostgreSQL · MongoDB

- ▶ Designed RBAC system enforced at both route middleware and data-access layer, supporting admin, organiser, and buyer roles. Dual-layer enforcement prevents privilege escalation via direct service calls — route-only RBAC does not.

EDUCATION

B.Tech, Textile Technology

Jul 2015 – May 2019

PSG College of Technology · Coimbatore, TN

Non-CS background. Self-directed transition via production backend projects, DSA training in Java/Spring Boot, and hands-on AWS cloud engineering.

CERTIFICATIONS & TRAINING

- ▶ **DSA in Java + Spring Boot Backend Specialisation** — Completion Certificate — Data structures, algorithms, Spring & Spring Boot

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